INSTRUCTOR-LED · HANDS-ON · PERSONALISED TO YOUR TEAM

Production AI & RAG for engineering teams

AI demos can look convincing while real requests expose missing evidence, inconsistent answers and costly rework. Coach your existing engineers to find the failure, measure it and improve it - using a RAG use case close to the product you want to build.

ON-SITE FORMAT	HANDS-ON GROUPS	CORE	STACK
2 full days on site	**Five to six engineers**	**RAG · Agents · Evals**	**Python · Azure**

TECHNOLOGIES AND TOPICS COVERED

Evaluations · **RAG** · **AI agents** · **Azure** · LLM fundamentals · Structured outputs · PDFs, tables & images · AI metadata & entities · Golden datasets · Feature / persona / scenario · Expert pass/fail review · Judge calibration · Error analysis · Chunking & embeddings · Hybrid search · Query routing · Clarification · Email knowledge · Contradictions · Tool calling · Trace annotation · State checks · MCP · Security & CI · Cost & latency

What each engineer can do afterwards

- Diagnose whether a failure comes from data, retrieval, prompting or the workflow.
- Create source-backed datasets and review coverage by feature, persona and scenario.
- Learn from support emails while keeping unanswered requests and contradictions out of the KB.
- Build a RAG pipeline and measure retrieval separately from answer quality.
- Calibrate an LLM judge against expert pass/fail reviews before trusting its scores.
- Evaluate an agent's answer, tool trace and actual state changes separately.

EACH ENGINEER LEARNS TO

- Turn a vague AI problem into a testable task and a measurable baseline.
- Run a repeatable cycle: inspect failures, test a change, measure the result.
- Apply the same method to new company data and use cases.

THE ORGANISATION GAINS

- AI skills inside the existing engineering team.
- Reusable foundations for more than one AI project.
- Evidence for decisions about quality, delivery effort and operating cost.

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For software engineers and technical leads. Working knowledge of Python and APIs.

Practice first: inspect company documents, evaluate a support system, then extend it with email knowledge and agent tools.

01 Foundations & company data

MODULES 01-03

01 module — AI & LLM fundamentals LAB

Training versus inference, tokens, context windows, sampling and model selection for quality, latency and cost. **Build:** Run a task and explain its limitations.

02 module — Prompting & structured outputs LAB

Instructions, few-shot examples, prompt chains, JSON schemas, validation and prompt versioning. **Build:** Make an unreliable output testable.

03 module — Documents & AI metadata LAB

Inspect PDF text, tables and images. Extract titles, summaries, categories, tags, product names, models and dates with evidence. **Build:** Compare the raw library with reviewed AI filters.

02 Datasets & trustworthy evaluation

MODULES 04-06

04 module — Golden datasets & coverage LAB

Define roles, situations and facts. Cover feature, persona and scenario; review question-answer pairs, source anchors and held-out splits. **Build:** Create a searchable dataset and find coverage gaps.

05 module — Expert review & success criteria LAB

A domain expert labels outcomes pass/fail and explains why. Separate bad answers from bad references; fix obvious failures first. **Build:** Annotate examples and turn critiques into a rubric.

06 module — LLM judges & calibration LAB

Build judge prompts from expert critiques. Separate prompt examples, development and test cases; inspect false passes, false failures, recall and specificity. **Build:** Tune on development cases; freeze and test the threshold.

03 RAG engineering

MODULES 07-09

07 module — RAG: the naive baseline LAB

Fixed chunks, overlap, embeddings, vector search and cited answers. Measure source precision/recall separately from correctness, faithfulness and completeness. **Build:** Run a baseline and inspect retrieved versus missing evidence.

08 module — RAG: chunks, retrieval & context LAB

Recursive and structure-aware chunks, sizes, tables, hybrid search, reranking, metadata filters and bounded parent context. **Build:** Compare retrieval changes on the same questions.

09 module — RAG: query flows & grounded answers LAB

History-to-query lists, intent routing, tailored prompts, source authority, citations and missing information. **Build:** Choose when to answer, clarify or abstain.

Production AI & RAG for engineering teams

Continue the V3 learning path: improve the system, learn from emails and evaluate agent actions.

04 Improvement, knowledge & agents

MODULES 10-14

10 module — Error analysis & experiments LAB

Group failures by category and cause. Test data, retrieval, prompt or workflow fixes; retain regressions and record cost. **Build:** Time-box one change and turn the failure into a regression check.

11 module — Email analysis & knowledge learning LAB

Classify requests, extract clarification fields, flag unanswered questions and compare replies with sources. Quarantine contradictions; admit reviewed lessons with provenance. **Build:** Reuse earlier email knowledge without leaking future answers.

12 module — Agents & operational tools LAB

Build the loop manually: call, observe, repeat. Search documents, check stock, calculate totals and create/edit sandbox orders; validate arguments and stop limits. **Build:** Inspect tool outputs and actual state changes.

13 module — Agent evaluations & trace annotation LAB

Evaluate answers, tool traces and state separately. Detect missing calls, wrong tools or parameters, bad scope and premature stops. **Build:** Annotate a step; test error detectors on positives and counterexamples.

14 module — Advanced agent use cases LAB

Real estate viewings, Instagram lead qualification and scheduling, ecommerce and travel. Each has tools, prompts, judges and scenario coverage. **Build:** Adapt one sandbox and test feature, persona and scenario combinations.

05 Secure production

MODULES 15-16

15 module — Security, orchestration & human review LAB

State graphs, persistence, retries, MCP, tool permissions, prompt injection and approval gates for actions. **Build:** Test recovery, duplicate requests and safe human handoffs.

16 module — Production, cost & Azure LAB

Load estimates, queues, concurrency, CI evaluations, logs, quality monitoring, latency, caching and cost budgets. **Build:** Deploy reusable infrastructure with operating limits.

Personalised to your team

A short skills quiz and project review tailor the modules and depth to your team.

Book a free call imosnoi@smartaitomation.com

How it is delivered

Two full days on site, personalised to your team and use case.

GROUP SIZE

Hands-on work in groups of five to six, so labs stay practical rather than becoming a demonstration.

VIRTUAL OPTION

4 half-days over 1 to 2 weeks, with the same content and labs.

LANGUAGE

Delivered in English, with pacing and written materials adjusted for cohorts working in a second language.

GETTING READY

A short technical readiness call and setup checklist are provided ahead of delivery, so the first morning starts with teaching rather than configuration.

About the trainer

Ion Moșnoi

AI/ML engineer · Production AI consultant · Engineering team coach

- **Production experience.** 11+ years in AI/ML and software engineering, with dozens of projects in production.
- **Research and publications.** Research in multi-agent coordination and computer vision in France and Finland. Published articles on transformer experiments for vision in 2019.
- **From startups to enterprises.** Fixed AI projects for companies including enterprises with 1B+ budgets for AI and digitalisation. Coaches engineers to build datasets, evaluate results and diagnose what to improve.

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